United International University (UIU)

Midterm Examination

IPE 401/IPE3401: Industrial Management/Industrial & Operational Management

Summer Trimester: 2025

Total time: 2 hours Total marks: 30

<u>There are 4 questions. You must answer any 3 questions</u>

1 **(a)** A project involves the following cash flows over the five years. The interest rate is 10% compounded annually. **[6]** **[CO2]**

Year	**0**	**1**	**2**	**3**	**4**	**5**
Cash Inflow ($)	0	3000	4000	x	8000	9500
Cash Outflow ($)	10000	2000	1000	9000	y	2000

If the future values of all net cash flows at the end of Year 5 are $867, calculate x and y considering the following constraint:

$$x - y = 4500$$

(b) Mohiuddin invests 45,000 Taka at 17% compounded hourly, 60,000 Taka at 16% compounded weekly, and 100,000 Taka at 12% compounded monthly. His friend Tawfique invests the same total amount of money in a single scheme that offers an annual interest rate of $p\%$, compounded annually. After 3 years, both of them withdrew the same amount of money. Find the value of p, showing all necessary calculations. **[4]** **[CO1]**

2 **(a)** Two projects are given: **[5]** **[CO2]**

Project "A"

Year	0	1	2	3	4	5	6
Cash Flow	-50,000	15,000	16,500	18,300	27,500	25,700	22,300

Project "B"

Year	**0**	**1**	**2**	**3**	**4**	**5**	**6**
Cash Flow	-50,000	18,000	12,000	16,200	22,600	19,500	20,000

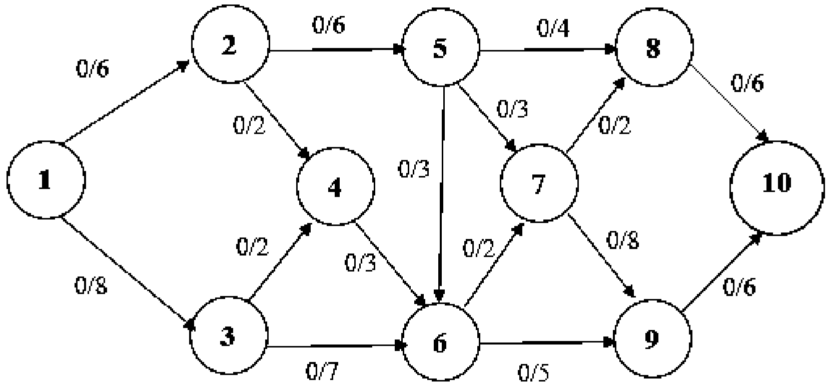
Using the Discounted Payback Period method, and assuming a nominal annual interest rate of **22.5%** compounded weekly, determine which project should be selected.

(b) **[5]** **[CO1]**

Year	**0**	**1**	**2**	**3**	**4**	**5**	**6**
Project 'S'	-75,000	20,300	20,500	24,900	23,970	31,700	24,300
Project 'L'	-90,000	24,700	22,500	28,850	24,980	33,700	19,300

Calculate the **IRR,** starting from **15%**, using the trial-and-error method. If the required rate of return is **20%**, should the company accept the projects? (You must show the necessary calculations).

3 (a) Find the maximum flow from the following diagram: **[5] [CO2]**



(b) Find the critical Path **[5] [CO1]**

Task	**Predecessor**	**Time (min)**
A	--	5
B	--	6
C	B	5
D	A	4
E	C, D	3
F	C	5
G	E	9
H	F	8
I	F	5

4 (a) A bicycle company uses 72,000 gearbox systems per year for its latest geared bicycle. The company manufactures them internally at a rate of 1,200 gearboxes per day. The cost of production of each gearbox is $20. During each production run, the setup cost is $100. The holding cost is 10% of the production cost of each gearbox. The company operates 300 days annually. Calculate the **optimal production quantity**, **total annual setup and holding cost,** and **cycle time**. **[4] [CO1]**

(b) A retailer sources men's shirts from a supplier who offers the following schedule: **[6] [CO2]**

Discount No.	**Order Quantity**	**Discount %**	**Unit Price ($)**
1	0 – 499	No discount	25
2	500 – 999	12	?
3	1000+	20	?

The retailer sells 10,000 shirts annually. The cost of ordering shirts each time is $75, while the holding cost is 20% of the unit price. Determine the **economic order quantity** and **associated unit cost**.

CO1-Apply Engineering economics and simple mathematics for Solving project selection problems for choosing the best possible project.
CO2- Analyze various industrial problems by using operation management, technique, operation research technique and cost accounting techniques and solve it.